BSc (Hons) in Computer Science

Year 3 Semester 1

SRS For RestroPlate Project

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Introduction

This Software Requirements Specification (SRS) provides a comprehensive description of the FoodShare Connect platform. The primary purpose of this system is to establish a sustainable ecosystem that connects surplus food donation providers, distribution centers, and consumers to reduce food waste and address food insecurity in local communities.

Aim

- Facilitate seamless connection between food donors (homes and restaurants) and distribution centers
- Enable real-time visibility of available food donations for consumers
- Provide efficient inventory management tools for distribution centers
- Reduce food waste and promote environmental sustainability
- Support community welfare by addressing food insecurity

Scope

- Donation providers (homes and restaurants)
    1. can register
    2. create donation listings,
    3. view donation requests from distribution centers,
    4. track their donation history

- Distribution centers can be managed
    1. donation requests,
    2. update inventory status,
    3. edit collected and distributed quantities,
    4. publish availability to consumers

- Consumers can view real-time availability of donations at nearby distribution centers without requiring authentication

Intended Audience

- Development Team
- Quality Assurance Team
- DevOps Team
- Project Managers
- System Architects

System Overview

Vision

FoodShare Connect envisions a world where no edible food goes to waste while community members face food insecurity. Our platform aims to create a transparent, efficient, and sustainable ecosystem that bridges the gap between food surplus and food need, fostering community solidarity and environmental responsibility.

Goal

- Reduce Food Waste
- Address Food Insecurity
- Streamline Operations.
  - Simplify the donation process for all stakeholders through intuitive digital tools.
- Enhance Transparency.
  - Provide real-time visibility of donation availability and distribution center inventory

Stakeholder

- Donation Providers
- Distribution Centers
- Consumers

Functional Requirements

1. **User Management (FR1)**
   - Registration and authentication for providers and centers.
   - **JWT-based** stateless authentication and **OAuth 2.0** social login.
   - Role-Based Access Control (RBAC) enforced at the API Gateway.

2. **Donation Management (FR2)**
   - Providers can create, modify, or cancel food listings.
   - Track donation status: *Available, Requested, Collected, Completed*.

3. **Distribution Center Operations (FR3)**
   - View and request available donations.

- Update inventory: Edit collected vs. distributed quantities.
- Toggle public visibility of items for consumers.

4. **Consumer Access (FR4)**
    - Real-time search and filter by proximity.
    - Public viewing of center profiles and available stock without login.

Non-Functional Requirements

- **Performance:** Real-time visibility via Redis; CDN for static asset distribution.
- **Scalability:** Horizontal scaling through Kubernetes and microservices.
- **Security:** Password hashing (bcrypt), input sanitization, and SSL/HTTPS enforcement.
- **Availability:** High availability of design on Azure cloud infrastructure

System Architectures

Use Case Diagram

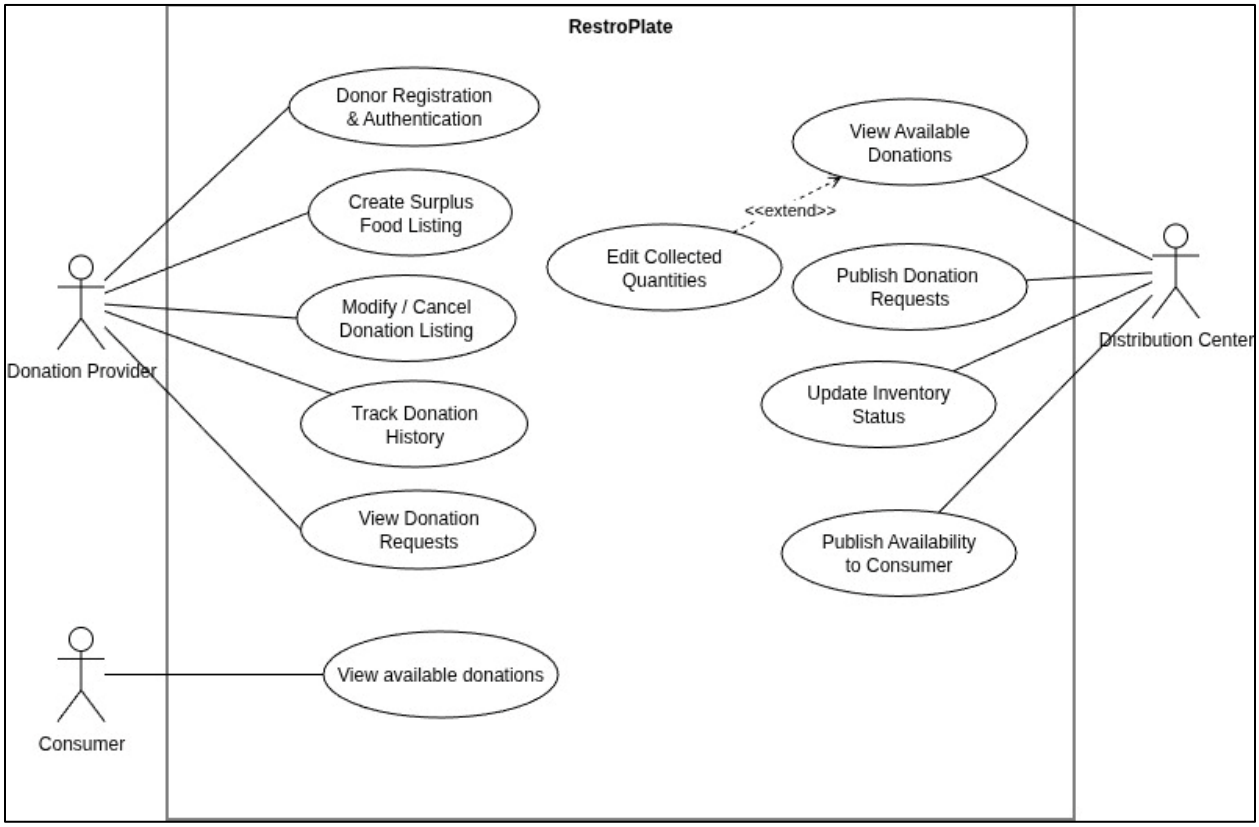

Figure 1: Use Case Diagram for RestroPlate

1. Presentation Layer

Responsive web interface built with the JavaScript framework React for donation providers, distribution centers, and consumers

2. Application Layer (Backend)

Core Modules (Service Boundaries):

- **User Module:** Manages user authentication, profile management, and role-based access control (RBAC).
- **Donation Module:** Handles the creation, modification, and retrieval of surplus food donation listings.

- **Distribution Module**: Manages distribution center operations, including inventory tracking and processing donation requests

3. Data Layer

- Primary Database: Use Superbase PostgreSQL database

EER Diagram

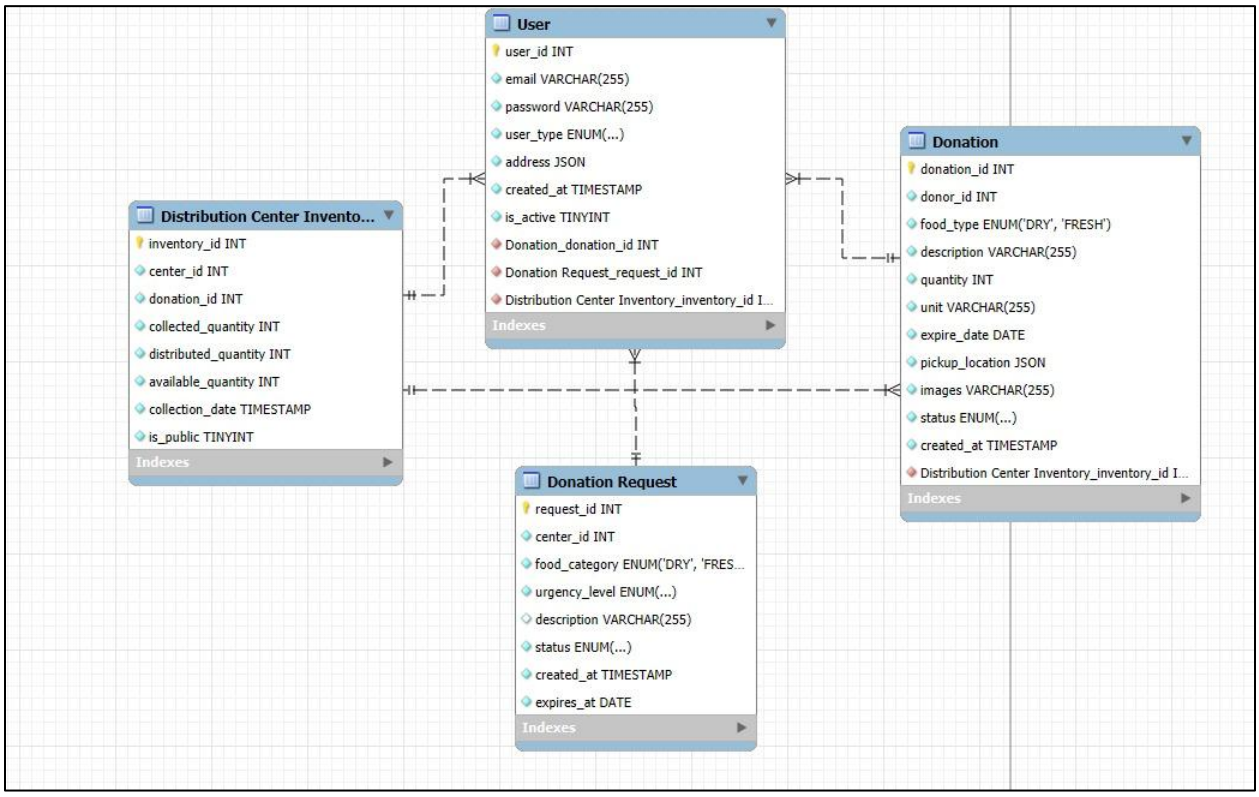

Figure 2: EER Diagram for RestroPlate

Infrastructure Layer

- **Container Orchestration:** Kubernetes for managing containerized microservices
- **CI/CD Pipeline:** Automated build, test, and deployment pipeline using GitHub Actions
- **CDN:** Content Delivery Network for static asset distribution
- **Cloud Provider:** Azure for scalable infrastructure